

あの震えるインタラクションを実装したい

2019-07-03 React LT会

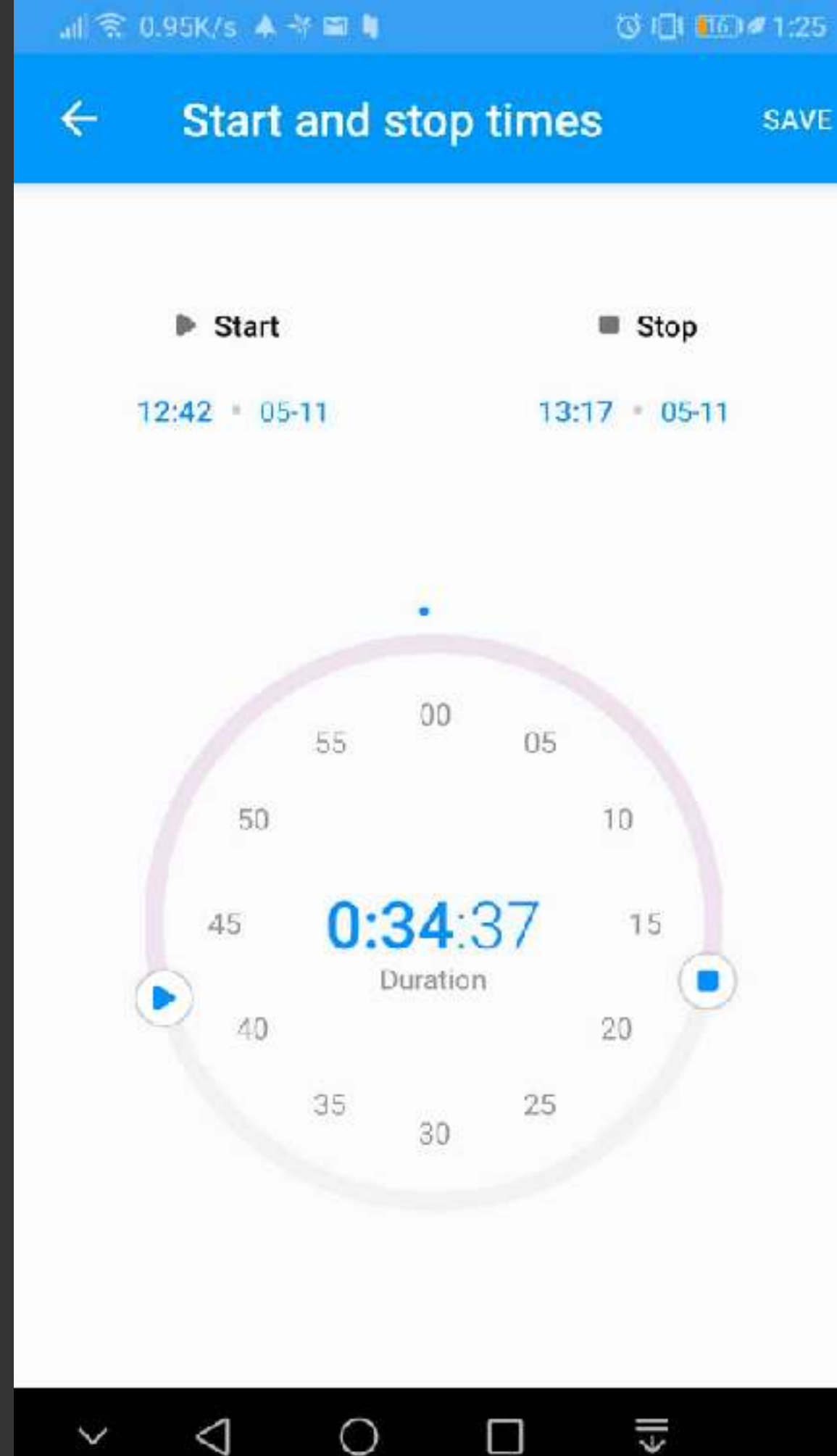
twitter: @nitaking\_

github: nitaking

# About

- **Satoshi Nitawaki**
- 日本酒好きエンジニア
- React Native, Node, TSなど
- 最近はOSS作ってみたり



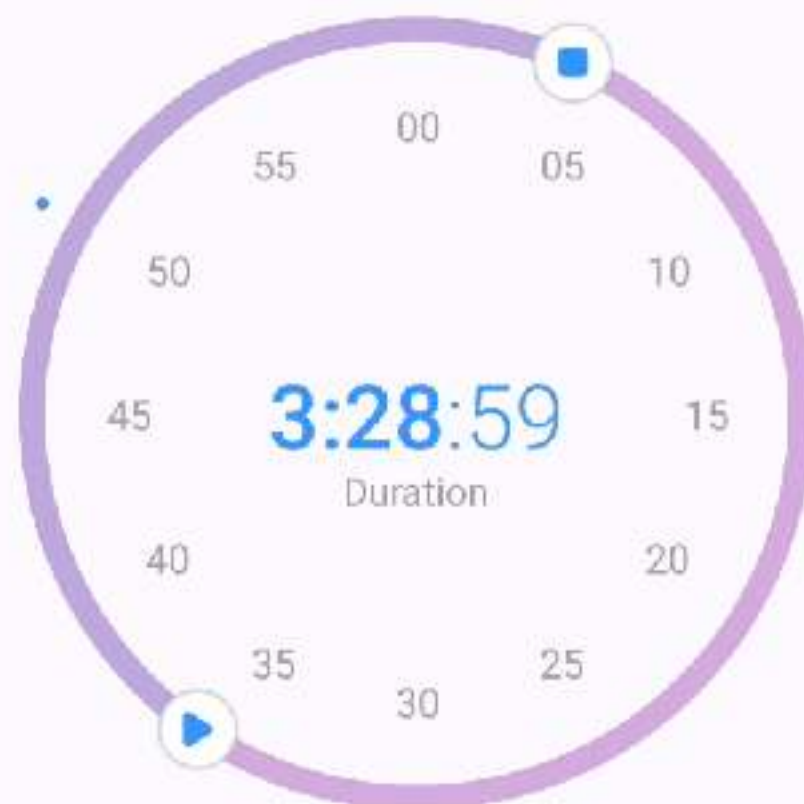




Start and stop times

SAVE

分が変更されるたびに  
小さなバンプが発生する



(iOSだとTwitterいいねボタン押したときの震えです)



これなに？

# Haptic (Haptic Feedback)

- 触覚フィードバック

# Haptic (Haptic Feedback)

iOS 10(iPhone7)からOSにAPI装備

iPhoneは日常的にポコポコ

→ iPhone 7 - 8のホームボタンもHaptic扱いかな

Androidはたまにしかポコポコ動かない

→ Androidユーザーには割と新鮮



# Haptic (Haptic Feedback)

ex)

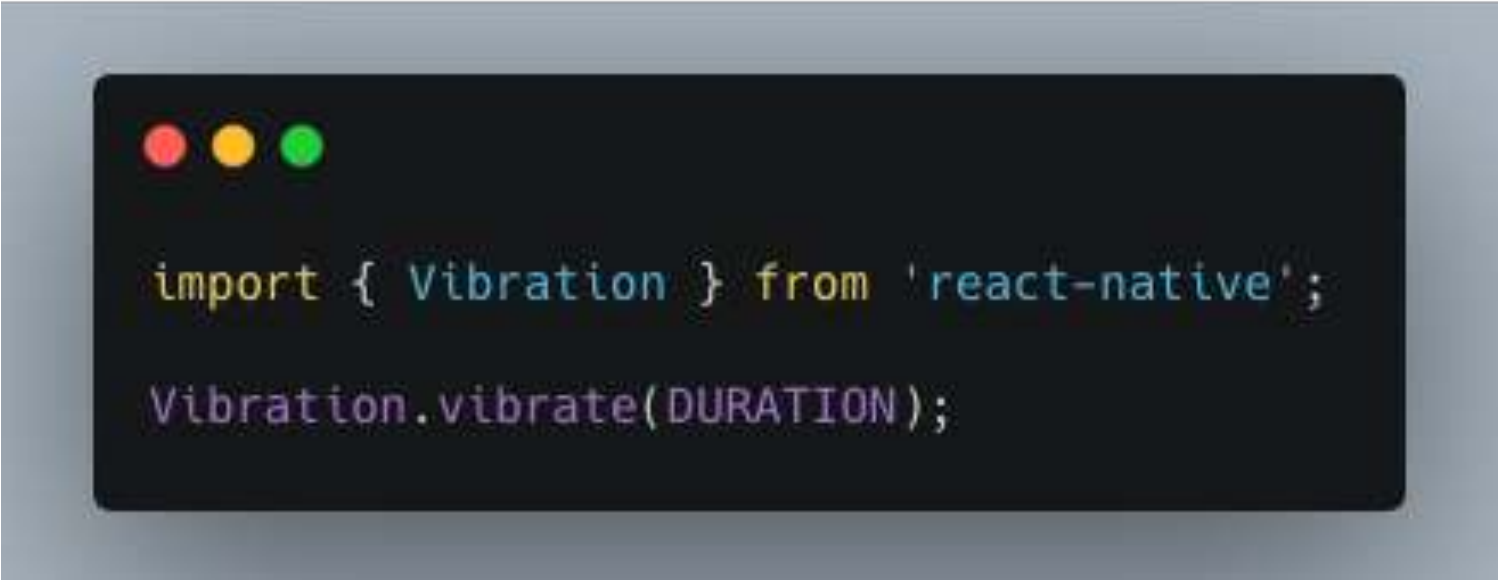
- togglの入力時間変更
- instagramのストーリー フィルター変更
- リスト画面のスワイプダウンして更新したとき (ios)
  - RNのRefreshControlのonRefresh時にiosは震えてたはず
- appstoreのインストール完了、失敗 (ios)
  - 基本的にiOSのエコシステム内は震えまくってるイメージ



React Nativeで実装してみたい

## 1. RNの標準Vibration

- バイブ時間短くしたら行けるかと思ったらそんなことはない
- 数msecセット→「ブーーン」
- これは絶対ちがう



```
import { Vibration } from 'react-native';  
Vibration.vibrate(DURATION);
```

## 2. expo-haptic

- 名前からして本命
- Expo APIのひとつ
- もちろんexpoアプリからはすぐに使用できる
- 他にHaptic系ライブラリは存在しなかった

A code editor window with a dark background and three colored window control buttons (red, yellow, green) in the top-left corner. It contains JavaScript code for using the expo-haptics library.

```
import * as Haptics from 'expo-haptics';  
  
Haptics.selectionAsync();  
Haptics.notificationAsync(type);  
Haptics.impactAsync(style);
```

## 2. expo-haptic

- Haptic.selectionAsync()
- Haptic.notificationAsync(NotificationFeedback)
  - success
  - warning
  - error
- Haptic.impactAsync(ImpactFeedback)
  - light
  - medium
  - heavy
- **Async表記のないmethodもありますがDeprecatedです**

素のRNで使えるんだっけ？

**react-native-unimodules**

# react-native-unimodules

- expo APIを**素のRNでも使用可能にする環境を提供**
- 導入の流れ
  1. `npm install react-native-unimodules`
  2. アプリの各機能へのPermission設定
    - • • unimoduleをinstallして使う



# 導入

- <https://github.com/unimodules/react-native-unimodules>

- **Android**

- android/settings.gradle 変更
- android/app/build.gradle 変更
- minSdkVersion : 16 → 21
- MainApplication.java 変更
- AndroidManifest.xml に  
permission追加

- 親切にDiffも用意されてるので  
導入はわかりやすい

## Configure Android

- At the top of `android/settings.gradle`, add `apply from: '../node_modules/react-native-unimodules/gradle.groovy'` and then on the next line add `includeUnimodulesProjects()`.
- Add `apply from: '../node_modules/react-native-unimodules/gradle.groovy'` anywhere in `android/app/build.gradle` and then `addUnimodulesDependencies()` inside `dependencies {}` block.
- If you need to customize the path to `node_modules`, for example because you are using yarn workspaces, then you can pass in a param `modulesPaths` for both of these functions: `includeUnimodulesProjects([modulesPaths: ['./path/to/node_modules']])`, `addUnimodulesDependencies([modulesPaths: ['./path/to/node_modules']])`
- If you need to exclude some of the unimodules that you are not using but they got installed by your other dependencies (like `expo`), then you can pass in `exclude` param for this. For example, if you want to exclude `expo-face-detector`, you may want to use this: `addUnimodulesDependencies([exclude: ['expo-face-detector']])`
- You can also customize the configuration of the unimodules dependencies (the default is `implementation`, if you're using Gradle older than 3.0, you will need to set `configuration: "compile"` in `addUnimodulesDependencies`, like: `addUnimodulesDependencies([configuration: "compile"])`
- We recommend using Java 1.8, you can set this in `android/app/build.gradle` like this.
- Update `minSdkVersion` in `android/build.gradle` to 21.
- Update your `MainApplication.java` to according to [this diff](#).

## Add permissions to AndroidManifest.xml

Add whichever of the following permissions you would like to use in your app to `android/app/src/main`:

```
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.CAMERA" />
<uses-permission android:name="android.permission.READ_CONTACTS" />
<uses-permission android:name="android.permission.READ_CALENDAR" />
<uses-permission android:name="android.permission.WRITE_CALENDAR" />
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.RECORD_AUDIO" />
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE" />
<uses-permission android:name="android.permission.WRITE_SETTINGS" />
```

tsapeta revised this gist on 26 Apr.

1 changed file with 8

```
1  import com.facebook.react.ReactActivity;
2  import com.facebook.react.shell.MainReactPackage;
3  import com.facebook.soloader.Soloader;
4  import com.mypapp.generated.BaseReactActivity;
5  import org.unimodules.adapters.react.ReactModuleRegistryAdapter;
6  import org.unimodules.adapters.react.ReactModuleRegistryProvider;
7  import org.unimodules.core.interfaces.SingletonModule;
8
9
10
11  import java.util.Arrays;
12  import java.util.List;
```



# 導入

- <https://github.com/unimodules/react-native-unimodules>
- **iOS**
  - Podfile 変更
  - AppDelegate.h 変更
  - AppDelegate.m 変更
  - Info.plist にpermission追加

# react-native-unimodules

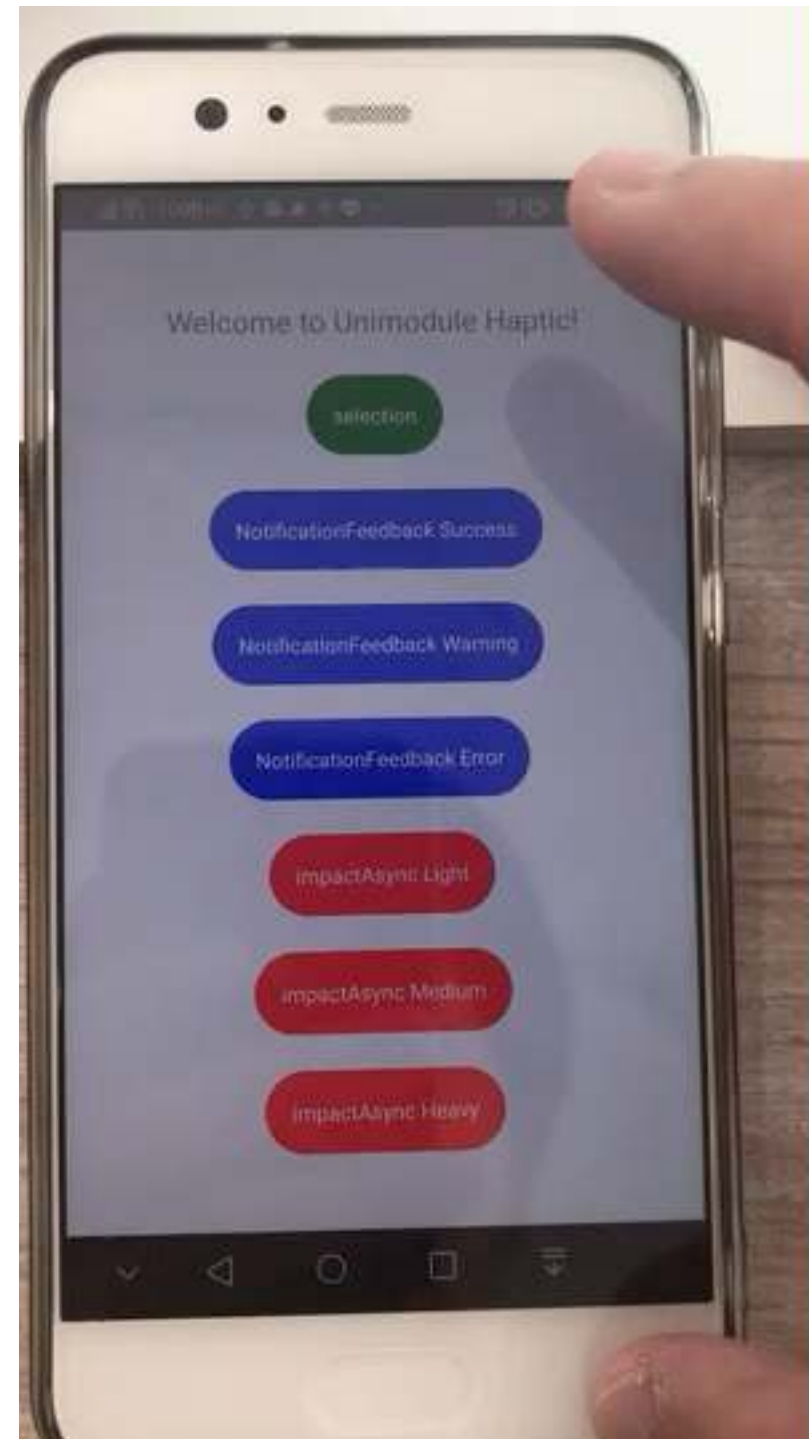
- react-native-unimodulesのpod install時
- ↓defaultで20個近く unimoduleがinstallされちゃう

アプリサイズを意識するなら個別にignore指定をしたほうがいい。（公式参照）

```
~/c/h/ios (master|+3) $ pod install
Ignoring ffi-1.9.18 because its extensions are not built. Try: gem pristine ffi --version 1.9.18
Ignoring nokogiri-1.8.0 because its extensions are not built. Try: gem pristine nokogiri --version 1.8.0
Ignoring unf_ext-0.0.7.5 because its extensions are not built. Try: gem pristine unf_ext --version 0.0.7.5
Installing unimodules:
expo-app-loader-provider@5.0.1 from ../node_modules/expo-app-loader-provider/ios
expo-constants@5.0.1 from ../node_modules/expo-constants/ios
expo-file-system@5.0.1 from ../node_modules/expo-file-system/ios
expo-haptics@5.0.1 from ../node_modules/expo-haptics/ios
expo-permissions@5.0.1 from ../node_modules/expo-permissions/ios
unimodules-barcode-scanner-interface@2.0.1 from ../node_modules/unimodules-barcode-scanner-interface/ios
unimodules-camera-interface@2.0.1 from ../node_modules/unimodules-camera-interface/ios
unimodules-constants-interface@2.0.1 from ../node_modules/unimodules-constants-interface/ios
unimodules-core@2.0.1 from ../node_modules/@unimodules/core/ios
unimodules-face-detector-interface@2.0.1 from ../node_modules/unimodules-face-detector-interface/ios
unimodules-file-system-interface@2.0.1 from ../node_modules/unimodules-file-system-interface/ios
unimodules-font-interface@2.0.1 from ../node_modules/unimodules-font-interface/ios
unimodules-image-loader-interface@2.0.1 from ../node_modules/unimodules-image-loader-interface/ios
unimodules-permissions-interface@2.0.1 from ../node_modules/unimodules-permissions-interface/ios
unimodules-react-native-adapter@2.0.1 from ../node_modules/@unimodules/react-native-adapter/ios
unimodules-sensors-interface@2.0.1 from ../node_modules/unimodules-sensors-interface/ios
unimodules-task-manager-interface@2.0.1 from ../node_modules/unimodules-task-manager-interface/ios
```

# Expo-haptics

- 素のRNにexpo-haptics入れてみた
- 音声がないとまったく伝わらないので  
後ほどツイッターにアップします！



他のunimoduleってどんなのがある？



# unimodules

<https://docs.expo.io/versions/latest/bare/unimodules-full-list/>

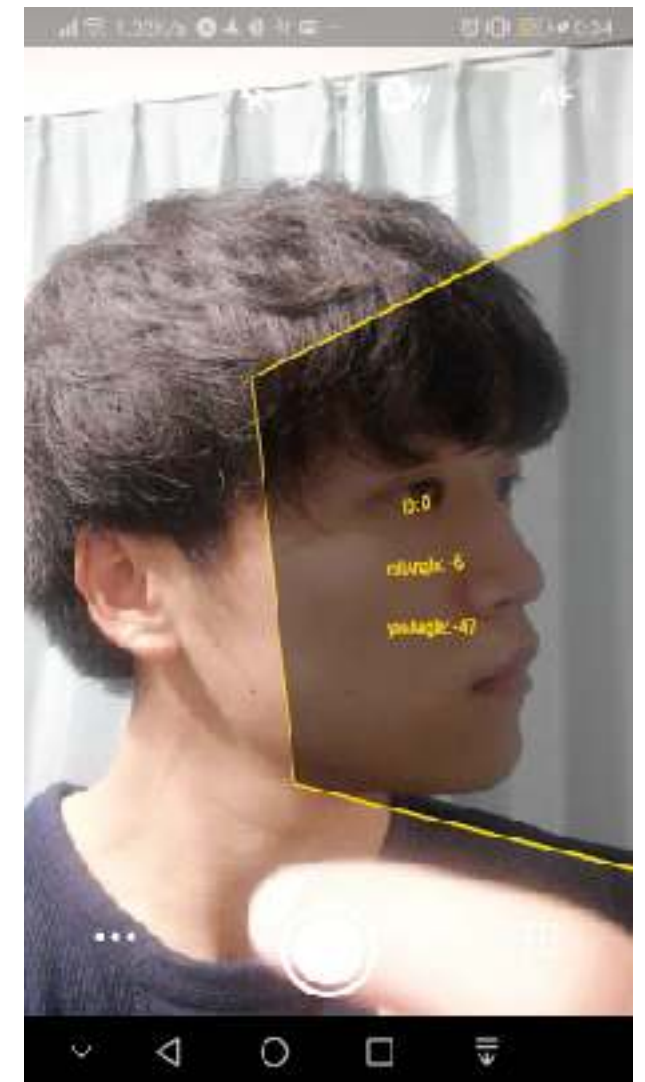
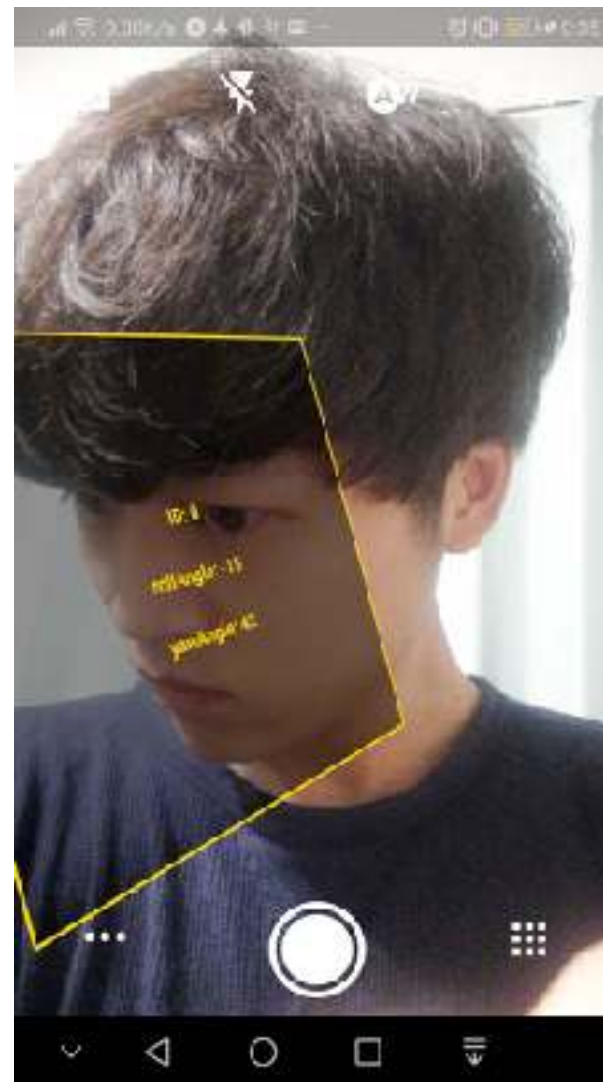
- expo-ads-admob
- expo-ads-facebook
- expo-analytics-amplitude
- expo-analytics-segment
- expo-app-auth
- expo-app-loader-provider
- expo-asset
- expo-av
- expo-background-fetch
- expo-barcode-scanner
- expo-blur
- expo-brightness
- expo-calendar
- expo-camera
- expo-constants
- expo-contacts-flutter-plugin
- expo-contacts
- expo-crypto
- expo-document-picker
- expo-face-detector
- expo-facebook
- expo-file-system
- expo-flutter-adapter
- expo-font
- expo-gl-cpp
- expo-gl
- expo-google-sign-in
- **expo-haptics**
- expo-image-manipulator
- expo-image-picker
- expo-in-app-purchases
- expo-intent-launcher
- expo-keep-awake
- expo-linear-gradient
- expo-local-authentication
- expo-localization
- expo-location
- expo-mail-composer
- expo-media-library
- expo-module-scripts
- expo-module-template
- expo-payments-stripe
- expo-permissions-flutter-plugin
- expo-permissions
- expo-print
- expo-processing
- expo-random
- expo-secure-store
- expo-sensors-flutter-plugin
- ...+10個くらい

# Expo APIs

- expo-ads-admob
- expo-ads-facebook
- expo-analytics-amplitude
- expo-analytics-segment
- expo-app-auth
- expo-app-loader-provider
- expo-asset
- expo-av
- expo-background-fetch
- expo-barcode-scanner
- expo-blur
- expo-brightness
- expo-calendar
- expo-camera
- expo-constants
- expo-contacts-flutter-plugin
- expo-contacts
- expo-crypto
- expo-document-picker
- expo-face-detector
- expo-facebook
- expo-file-system
- expo-flutter-adapter
- expo-font
- expo-gl-cpp
- expo-gl
- expo-google-sign-in
- **expo-haptics**
- expo-image-manipulator
- expo-image-picker
- expo-in-app-purchases
- expo-intent-launcher
- expo-keep-awake
- expo-linear-gradient
- expo-local-authentication
- expo-localization
- expo-location
- expo-mail-composer
- expo-media-library
- expo-module-scripts
- expo-module-template
- expo-payments-stripe
- expo-permissions-flutter-plugin
- expo-permissions
- expo-print
- expo-processing
- expo-random
- expo-secure-store
- expo-sensors-flutter-plugin
- ...+10個くらい

# 例えば

- expo-face-detector
  - 顔認識
    - 謎精度
  - 微笑み率とれる
  - 目を開いている率も
- **正面OK,横向き写真NG**  
**のバリデーション**  
**に使える？**



expoは素RNに  
関係のない話だと思っていた



expoの優秀なunimodulesは  
実は素のRNでも使える

Enjoy RN dev! 🎉